

Scalable watermarking for identifying large language model outputs

Dathathri et al., *Nature* (2024)

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QuantCo reading group

7 Oct 2026

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Motivation

The Guardian UK

News Opinion Sport Culture Lifestyle

World Europe US news Americas Asia Australia Middle East Africa Inequality Global development

AI (artificial intelligence)

This article is more than 2 months old

AI labels to be compulsory on authentic-looking content under EU rules

Companies must ensure people know when they are interacting with artificially generated images, audio and text designed to look real

Jennifer Rankin in Brussels
Fri 31 Jul 2026 12:21 BST

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Screenshot of an AI-generated image of the French president, Emmanuel Macron, dancing in a 1980s nightclub.

Advertisement

Could you pay less for your pension?

Capital at risk.

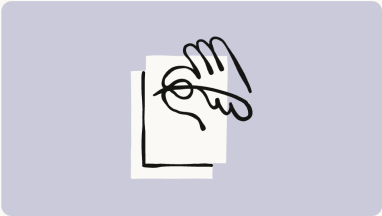
[Link.](#) EU AI Act enforcement, starting from 2nd August 2026.

ResearchPolicyCommitmentsLearnNewsTry Claude

Announcements

How Claude's text watermark works

Aug 14, 2026



What is watermarking?

How does watermarking affect Claude's outputs?

Which specific method of watermarking do you use?

What about cases where Claude has profited or misled human users?

Future Claude models will generate text that contains a watermark. This is a way of determining the likelihood that Claude was involved in writing the text, and we, along with several other major AI providers, are implementing this change to comply with the EU AI Act.

In this article, we share answers to some of the questions we've received about how our chosen watermarking method works, whether it affects Claude's outputs, and why we're making this change. To summarize:

[Link.](#) Anthropic adopts text watermarking.

Writing in Acceptance Criteria



Transactions on Machine Learning Research

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2 min read · Aug 28, 2026



9



Three things are true today:

1. Peer reviews generally put least emphasis on the criterion of clear writing. From our experience, this is true for TMLR as well.
2. Journals and conferences including TMLR are seeing a large number of submissions, including potentially AI generated or heavily AI written ones.
3. AI writing is currently often convoluted, has unnecessary jargon, and not easy to understand.

[Link](#). TMLR's call for reviewers to emphasise more on clarity of writing due to surges in AI-generated content.

Background

How LLMs Generate Texts

- Most LLMs generate text using autoregressive next-token sampling.



- Example:

	Vocabulary	p_{LM}		Selected token
... my favourite tropical fruit is	mango	0.50	→	lychee
	lychee	0.30		
	papaya	0.15		
	durian	0.05		

Existing Approaches for LLM Text Detection



Retrieval-based detection

- Keep a growing record and check against it for matches.
- Requires scale and coordination; raises privacy concerns.



Post hoc detection

- Use the statistical features of text or train an ML-based classifier.
- Computationally expensive; inconsistent performance (especially out-of-domain)



Watermarking

- Mark generated text so that it can be subsequently identified.
- Quality (of generated texts) and effect (of detection) trade-off.
- Three approaches:
 - During generation.
 - Editing already generated text.
 - Altering LLM training data.

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← This paper

Intuition behind watermarking – Monopoly



Method 1: roll a dice

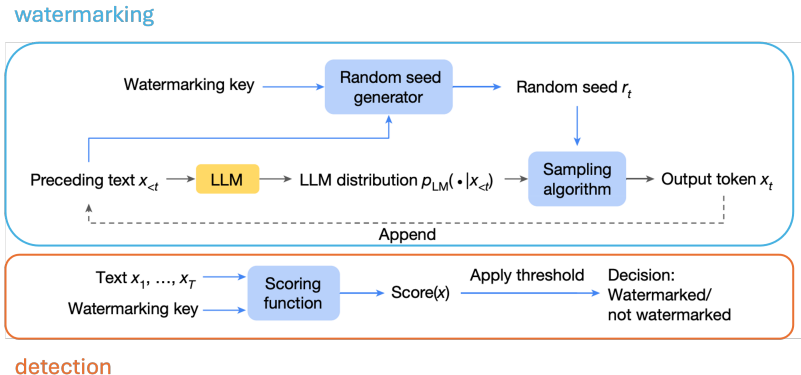
π Method 2: book of π (3.14159265358979323846...)

Both are random, but Method 2 is back-traceable (watermarked).

SynthID-Text

- <https://github.com/google-deepmind/synthid-text>
- <https://watermark.keito.me/>

SynthID-Text overview

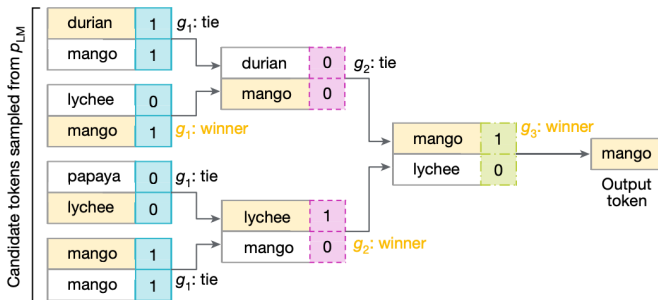


- Non-distortionary.
- Computationally efficient.

Fig. 1, Dathathri et al.

Watermarking – Tournament Sampling

Tournament Sampling: use a tournament-like process to select a token from the LLM distribution that is likely to score higher under some random watermarking functions.



Watermarking function: $g_l(x_t, r_t)$, where x_t is a candidate token, and r_t is a random seed.

- Number of layers m . Larger $m \rightarrow$ better detectability. They use $m = 30$.
- Number of candidates N . Larger $N \rightarrow$ better detectability. They use $N = 2$.

Watermarking function

Example (binary watermarking function)

$g_l(x, r) = \mathbb{1}\{u_l(x, r) > 0.5\}$, where

$$u_l(x, r) = \frac{h(x, r, l)}{2^{n_{\text{sec}}}} \in [0, 1) ,$$

h is a pseudorandom function producing uniform-looking integers, and n_{sec} is the security parameter.

Watermarking detection (Appendix A)

$$\text{Score}(x) = \frac{1}{mT} \sum_{t=1}^T \sum_{l=1}^m g_l(x_t, r_t) .$$

Intuition: $\text{Score}(x)$ takes higher values for a watermarked sequence.

Context index t is only summed over a subset $\hat{T} \subseteq \{1, \dots, T\}$ that

- excludes the first $t = 1, \dots, H$ tokens, where the context window is incomplete.
- excludes t where the context $x_{t-H}, \dots, x_{t-1}$ has appeared previously in the sequence.

Example ($H = 3$)

"Explain in one short paragraph how to come up with nice banters for machine learning talks tailored
incomplete
for machine learning experts."
repeated

Watermarking detection (Appendix A)

Hypothesis testing for watermarking

Under

H_0 : the text is not watermarked by SynthID-Text ,

each $g_l(x_t, r_t)$ is i.i.d., and the p -value of $\text{Score}(x)$ has an explicit form.

Quality preservation (Appendix G)

In words

A sampling algorithm is non-distortionary if in expectation the distribution on the next token is identical before and after watermarking.

Definition 16 (Single-token non-distortionary sampling algorithm). *A sampling algorithm $\mathcal{S} : \Delta V \times \mathcal{R} \rightarrow V$ is (single-token) non-distortionary if for any probability distribution $p \in \Delta V$ and token $x \in V$:*

$$\mathbb{E}_{r \sim \text{Unif}(\mathcal{R})} [\mathbb{P}(\mathcal{S}(p, r) = x)] = p(x).$$

- V = set of candidate tokens.
- ΔV = probability simplex over $V = \{p : V \rightarrow [0, 1] \mid \sum_{x \in V} p(x) = 1\}$.
- $\mathcal{R}$ = set of possible random seeds.

Quality preservation (Appendix G)

K -sequence non-distortion: a stronger version of single-token non-distortion that holds for **sequences** of tokens.

Theorem 21

For any $K \geq 1$, SynthID-Text (with $N = 2$ and K -sequence repeated context masking) is K -sequence non-distortionary.

Example (K -sequence repeated context masking, $K = 2$)

"Explain in one short paragraph how to come up with nice banters for machine learning talks tailored **for machine learning** experts."

"Give me a concrete example of a joke **for machine learning** that I can use to start my talk."

Computational complexity

Sampling method	Latency (ms/token)	Increase (%)
Baseline	15.527	–
Tournament sampling ($m = 30$ layers)	15.615	0.57
Gumbel sampling	–	0.26
Soft Red List	–	0.28

Table: Gemma 7B-IT latency on 4 v5e TPUs.

SynthID-Text can be integrated into large-scale production systems with “negligible additional computational overhead”.

Empirical evaluation

Empirical evaluation – scalability to number of tokens

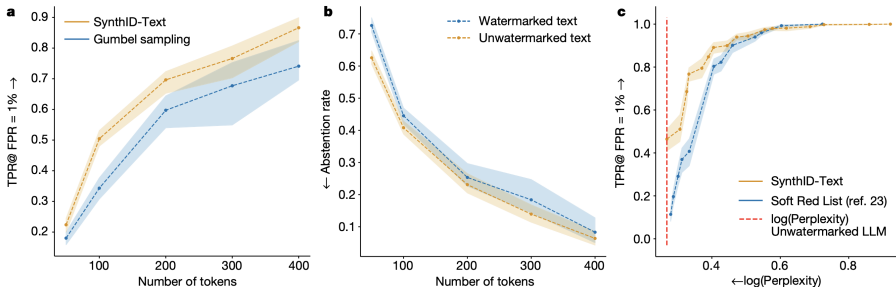
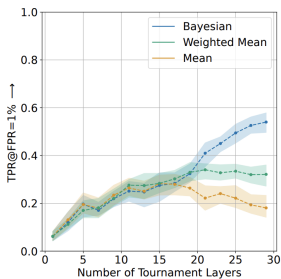
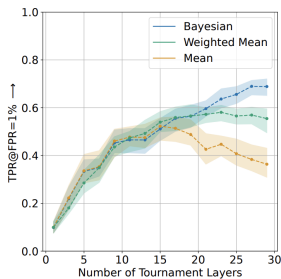


Fig.3: Detection performance of SynthID-Text, using prompts from the ELI5 dataset.

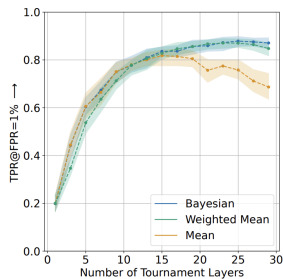
Empirical evaluation – scalability to m



(a) Temperature=0.5



(b) Temperature=0.7



(c) Temperature=1.0

Fig.C1: Detectability with respect to number of tournament layers m .

Empirical evaluation – robustness under perturbation

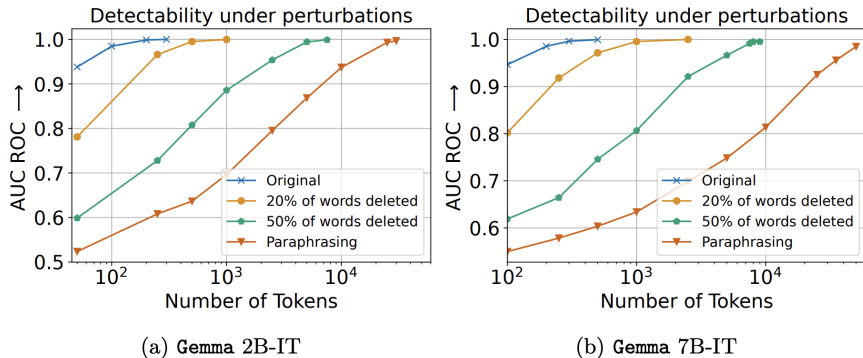


Fig.C3: Detectability under various perturbations.

Discussion

Some remarks

- **Diversity**: Same diversity within a single text, but reduced diversity for the same prompt.
- **Limitation**: Requires full knowledge of the watermarking scheme for the detection to work —
 - ▶ NOT a generic LLM-content detector.
 - ▶ Alternatives exist: pangram.com
- **Adversarial attacks**: Even a simple translation trick can be effective.
- **Evaluation**: Various evaluation principles presented (frequentist, Bayesian, with/without perturbation). What is EU's rule?